



Vertical Asymptotes

What Breaks This?

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* The Rule // What BREAKS This?

- **Vertical Asymptote**

Set the **denominator** equal to zero and solve.

The VA is a **forbidden input** — the function explodes there!

Answer is always: $x = \text{---}$ (because it's an input)

* Find the VA

$$f(x) = \frac{3}{x-5}$$

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Set denominator = 0: $x - 5 = 0$

$$\text{VA: } x = 5$$

* Find the VA

$$f(x) = \frac{x+1}{x+4}$$

* Find the VA

$$f(x) = \frac{x+1}{x+4}$$

Set denominator = 0: $x + 4 = 0$

VA: $x = -4$

* Find the VA

$$f(x) = \frac{2x}{x-7}$$

* Find the VA

$$f(x) = \frac{2x}{x-7}$$

Set denominator = 0: $x - 7 = 0$

VA: $x = 7$

* Find the VA

$$f(x) = \frac{1}{x}$$

* Find the VA

$$f(x) = \frac{1}{x}$$

Set denominator = 0: $x = 0$

VA: $x = 0$

- **Pro Tip**

This is the parent function! VA at the y -axis.

* Find the VA // Solve a 2-Step

$$f(x) = \frac{5x + 25}{2x + 4}$$

* Find the VA // Solve a 2-Step

$$f(x) = \frac{5x + 25}{2x + 4}$$

Set denominator = 0: $2x + 4 = 0$

$$2x = -4 \rightarrow x = -2$$

VA: $x = -2$

* Find the VA // Solve a 2-Step

$$f(x) = \frac{x - 1}{3x - 9}$$

* Find the VA // Solve a 2-Step

$$f(x) = \frac{x - 1}{3x - 9}$$

Set denominator = 0: $3x - 9 = 0$

$$3x = 9 \rightarrow x = 3$$

VA: $x = 3$

* Find the VA // Two Asymptotes!

Watch Out!

If the denominator factors into **two** pieces, you get **two VAs**!

$$f(x) = \frac{-4}{x^2 - 9}$$

* Find the VA // Two Asymptotes!

Watch Out!

If the denominator factors into **two** pieces, you get **two VAs**!

$$f(x) = \frac{-4}{x^2 - 9}$$

Factor: $x^2 - 9 = (x - 3)(x + 3)$

$$x - 3 = 0 \rightarrow x = 3 \quad x + 3 = 0 \rightarrow x = -3$$

VA: $x = 3$ and $x = -3$

* Find the VA // Two Asymptotes!

$$f(x) = \frac{1}{x^2 - 4}$$

* Find the VA // Two Asymptotes!

$$f(x) = \frac{1}{x^2 - 4}$$

Factor: $x^2 - 4 = (x - 2)(x + 2)$

$$x - 2 = 0 \rightarrow x = 2 \quad x + 2 = 0 \rightarrow x = -2$$

VA: $x = 2$ and $x = -2$

* Find the VA // Two Asymptotes!

$$f(x) = \frac{x+1}{x^2+x-6}$$

* Find the VA // Two Asymptotes!

$$f(x) = \frac{x+1}{x^2+x-6}$$

Factor: $x^2 + x - 6 = (x+3)(x-2)$

$$x+3=0 \rightarrow x=-3 \quad x-2=0 \rightarrow x=2$$

VA: $x = -3$ and $x = 2$

* Your Turn!

▷ YouTry!

Find the VA:

1. $f(x) = \frac{7}{x+6}$ VA: _____

2. $f(x) = \frac{x}{2x-10}$ VA: _____

3. $f(x) = \frac{3x}{x^2-16}$ VA: _____

* Your Turn!

▷ YouTry!

Find the VA:

1. $f(x) = \frac{7}{x+6}$ VA: _____

2. $f(x) = \frac{x}{2x-10}$ VA: _____

3. $f(x) = \frac{3x}{x^2-16}$ VA: _____

1. $x = -6$

2. $x = 5$

3. $x = 4$ and $x = -4$



Go Sandies!



Nothing breaks a Sandie.